Junko Munakata Marr, Professor

Computer Science

HES Robotics Track Representatives

Estelle Smith, Assistant Professor

Thomas Williams, Associate Professor, Director of MIRROR Lab

Engineering, Design & Society

Alina Handorean, Teaching Professor (joint appointment: EDS/CEE)

Dean Nieusma, Associate Professor and Department Head

Elizabeth Reddy, Assistant Professor

Jessica Smith, Professor

Juan Lucena, Professor and Director, Humanitarian Engineering Undergraduate Programs

Kevin Moore, Professor and Executive Director of Humanitarian Engineering Undergraduate Programs

Marie Stettler Kleine, Assistant Professor

Geology and Geological Engineering

HES Geology Track Representative

Paul Santi, Professor

Geophysics

HES Geophysics Track Representatives

Richard Krahenbuhl, Associate Research Professor, Director of Humanitarian Engineering and Science Graduate Program

Jeffrey Shragge, Professor

Hydrologic Science and Engineering

Program Description

Hydrologic Science and Engineering (HSE) is comprised of faculty from several different Mines departments and offers interdisciplinary graduate degrees in hydrology.

The program offers programs of study in fundamental hydrologic science and applied hydrology with engineering applications. Our program encompasses groundwater hydrology, surface-water hydrology, vadose-zone hydrology, watershed hydrology, contaminant transport and fate, contaminant remediation, hydrogeophysics, and water policy/law.

HSE requires a core study of formal graduate courses for all degrees. Programs of study are interdisciplinary in nature, and coursework is obtained from multiple departments at Mines and is approved for each student by the student's advisor and thesis committee.

Primary Contact

Program Director: Eric Anderson
Program Coordinator: Rachel McDonald

https://hydrology.mines.edu/

Program Requirements

Prerequisites

- Baccalaureate degree in a science or engineering discipline
- College calculus: two semesters required
- Differential equations: one semester required
- College physics: one semester required
- College chemistry: two semesters required
- College statistics: one semester required
- Fluid mechanics

Note that some prerequisites may be completed in the first few semesters of the graduate program if approved by the HSE director/program manager. Contact Rachel McDonald for questions at rmcdonald@mines.edu.

Mines' Combined Undergraduate/Graduate Degree Program

Students enrolled in Mines' combined undergraduate/graduate program may double count up to 6 credits of graduate coursework to fulfill requirements of both their undergraduate and graduate degree programs. These courses must have been passed with B- or better, not be substitutes for required coursework, and meet all other university, department, and program requirements for graduate credit.

Students are advised to consult with their undergraduate and graduate advisors for appropriate courses to double count upon admission to the combined program.

Masters Non-Thesis

Core Curriculum

Curriculum areas of emphasis consist of core courses, design/thesis, and electives. Core courses cover four areas of knowledge: Groundwater, Surface Water, Chemistry, and Contaminant Fate and Transport. Students take 12 credits of core courses as listed in Table 1. A grade of B- or better is required in all core courses for graduation.

Table 1

GEGN566	GROUNDWATER ENGINEERING	3.0
GEGN582	INTEGRATED SURFACE WATER HYDROLOGY	3.0
CEEN550	PRINCIPLES OF ENVIRONMENTAL CHEMISTRY	3.0
CEEN580	CHEMICAL FATE AND TRANSPORT IN THE ENVIRONMENT	3.0
Design Course (Table 2) or Independent Study		3.0
Electives (Table 3)		15.0
Total Semester Hrs		**30.0**

Students who have completed coursework for a previous degree that satisfies one of these requirements can get core curriculum requirements waived with the appropriate Waiver form and approval of advisor.

In addition, a fluid mechanics class is required for students to complete the HSE degree programs. If a student has previously taken a fluid mechanics course (for example as part of an undergraduate degree) then this requirement is met; if a student has not previously taken a fluid

mechanics course this requirement can be satisfied by taking: GEGN/CEEN 585 – Fluid Mechanics for Hydrology.

Design Courses

For non-thesis MS students, they must choose at least one of the Design Courses listed in Table 2. Students may complete an independent study course in lieu of the design course.

Table 2

CEEN515	HILLSLOPE HYDROLOGY AND STABILITY	3.0
CEEN581	WATERSHED SYSTEMS MODELING	3.0
CEEN575	HAZARDOUS WASTE SITE REMEDIATION	3.0
CEEN584	SUBSURFACE CONTAMINANT TRANSPORT	3.0
GEGN532	GEOLOGICAL DATA ANALYSIS	3.0
GEGN575	APPLICATIONS OF GEOGRAPHIC INFORMATION SYSTEMS	3.0
GEGN583	MATHEMATICAL MODELING OF GROUNDWATER SYSTEMS	3.0
GEGN584	FIELD METHODS IN HYDROLOGY	3.0
GEGN586	NUMERICAL MODELING OF GEOCHEMICAL SYSTEMS	3.0

Elective Courses

Electives may be chosen from the approved list below or as approved by your advisor or thesis committee.

Table 3

CEEN511	UNSATURATED SOIL MECHANICS	3.0
CEEN512	SOIL BEHAVIOR	3.0
CEEN515	HILLSLOPE HYDROLOGY AND STABILITY	3.0
CEEN555	LIMNOLOGY	3.0
CEEN560	MOLECULAR MICROBIAL ECOLOGY AND THE ENVIRONMENT	3.0
CEEN562	ENVIRONMENTAL GEOMICROBIOLOGY	3.0
CEEN570	WATER AND WASTEWATER TREATMENT	3.0
CEEN571	ADVANCED WATER TREATMENT ENGINEERING AND WATER REUSE	3.0
CEEN575	HAZARDOUS WASTE SITE REMEDIATION	3.0
CEEN586	HYDROMETEOROLOGY	3.0
CEEN581	WATERSHED SYSTEMS MODELING	3.0
CEEN582	VADOSE ZONE HYDROLOGY	3.0
GEGN532	GEOLOGICAL DATA ANALYSIS	3.0
GEGN573	GEOLOGICAL ENGINEERING SITE INVESTIGATION	3.0
GEGN575	APPLICATIONS OF GEOGRAPHIC INFORMATION SYSTEMS	3.0
GEGN581	ANALYTICAL HYDROLOGY	3.0
GEGN584	FIELD METHODS IN HYDROLOGY	3.0
GEGN586	NUMERICAL MODELING OF GEOCHEMICAL SYSTEMS	3.0
GEOL540	ISOTOPE GEOCHEMISTRY AND GEOCHRONOLOGY	3.0
MATH530	INTRODUCTION TO STATISTICAL METHODS	3.0
MATH531	THEORY OF LINEAR MODELS	3.0
MATH532	SPATIAL STATISTICS	3.0
EBGN510	NATURAL RESOURCE ECONOMICS	3.0
HASS588	GLOBAL WATER POLITICS AND POLICY	3.0
GEGN585	FLUID MECHANICS FOR HYDROLOGY	2.0

Thesis Degrees

MS thesis: 30 credits total, consisting of 24 credits of coursework and 6 credits of thesis credit. Students must also write and orally defend a research thesis.

PhD: 72 total credits, consisting of coursework (at least 36 hours), and research (at least 24 hours). Students must also successfully complete qualifying examinations, write and defend a dissertation proposal, write and defend a doctoral dissertation, and are expected to submit the dissertation work for publication in scholarly journals.

Required Curriculum

Students will work with their academic advisors and graduate thesis committees to establish plans of study that best fit their individual interests and goals. Each student will develop and submit a plan of study to their advisor during the first semester of enrollment. Doctoral students may transfer in credits from an earned MS graduate program according to requirements listed in the Graduate Degrees and Requirements (p. 29) section of the graduate bulletin, and after approval by the student's thesis committee.

Thesis and Dissertation Committee Requirements

Students must meet the general requirements listed in the graduate bulletin section *Graduate Degrees and Requirements*. In addition, the student's advisor or co-advisor must be an HSE faculty member. For MS thesis students, at least two committee members must be members of the HSE faculty. For doctoral students, at least two faculty on the committee must be a member of the HSE faculty. For PhD committee the required at-large member must be from a Mines department outside the student's home department, and where applicable, outside the students minor department.

Core Courses

Curriculum areas of emphasis consist of core courses, design/thesis, and electives. Core courses cover four areas of knowledge: Groundwater, Surface Water, Chemistry, and Contaminant Fate and Transport. Students take 9 credits of core courses as listed in Table 4. A grade of B- or better is required in all core courses for graduation.

Table 1

GEGN566	GROUNDWATER ENGINEERING	3.0
GEGN582	INTEGRATED SURFACE WATER HYDROLOGY	3.0
Choose at least one of the courses below		
CEEN550	PRINCIPLES OF ENVIRONMENTAL CHEMISTRY	3.0
CEEN580	CHEMICAL FATE AND TRANSPORT IN THE ENVIRONMENT	3.0

Students who have completed coursework for a previous degree that satisfies one of these requirements can get core curriculum requirements waived with the appropriate Waiver form and approval of advisor.

In addition, a fluid mechanics class is required for students to complete the HSE degree programs. If a student has previously taken a fluid mechanics course (for example as part of an undergraduate degree) then this requirement is met; if a student has not previously taken a fluid

mechanics course this requirement can be satisfied by taking: GEGN/CEEN 585 – Fluid Mechanics for Hydrology.

Electives

Elective courses may be chosen from the approved list in Table 3 or as approved by your advisor or thesis committee.

Areas of specialization

Students may choose to complete an area of specialization within the MS or PhD in Hydrology degrees by taking additional defined courses. These areas of specialization are: Hydrogeophysics, Hydrobiogeochemistry, and Hydrology, Policy, and Management. Courses required for these areas of specialization are:

1. Hydrogeophysics:

GPGN574	ADVANCED HYDROGEOPHYSICS	3.0
GPGN533	GEOPHYSICAL DATA INTEGRATION & GEOSTATISTICS	3.0
GPGN570	APPLICATIONS OF SATELLITE REMOTE SENSING	3.0
or GPGN520	ELECTRICAL AND ELECTROMAGNETIC EXPLORATION	

2. Hydrobiogeochemistry

Students choose three of the following courses with at least one from each of microbiology-focused and geochemistry focused courses. Students with a Hydrobiogeochemistry area of specialization are encouraged to enroll in CEEN550 and a separate Contaminant Fate and Transport course (CEEN580 or CEEN584) to satisfy the HSE core, leaving GEGN586 and CEEN551 as the geochemistry focused courses.

Microbiology Focus:		
CEEN562	ENVIRONMENTAL GEOMICROBIOLOGY	3.0
CEEN560	MOLECULAR MICROBIAL ECOLOGY AND THE ENVIRONMENT	3.0
Geochemistry Focus:		
CEEN550	PRINCIPLES OF ENVIRONMENTAL CHEMISTRY	3.0
GEGN586	NUMERICAL MODELING OF GEOCHEMICAL SYSTEMS	3.0
CEEN551	ENVIRONMENTAL ORGANIC CHEMISTRY	3.0

3. Hydrology, Policy, and Management

Students pursuing the Hydrology, Policy, and Management specialty track will choose two of the following three courses focused on water policy and management.

EBGN537	ECONOMICS OF WATER	3.0
HASS588	GLOBAL WATER POLITICS AND POLICY	3.0
HASS584	US WATER POLITICS AND POLICY	3.0

In addition, students will choose a third course from a broader list that also includes courses in complementary areas of communication, economics, law, philosophy, and policy. Current course options are listed below. Because course options are continually expanding, additional complementary courses (beyond those listed here) may be approved on an ad hoc basis by the coordinator of the Hydrology, Policy, and Management track and the HSE program director in response to individual student requests.

EBGN537	ECONOMICS OF WATER	3.0
EBGN570	ENVIRONMENTAL ECONOMICS	3.0
HASS521	ENVIRONMENTAL PHILOSOPHY	3.0
HASS523	ADVANCED SCIENCE COMMUNICATION	3.0
HASS525	ENVIRONMENTAL COMMUNICATION	3.0
HASS560	GEOPOLITICS OF NATURAL RESOURCES	3.0
HASS565	SCIENCE, TECHNOLOGY, AND SOCIETY	3.0
HASS568	ENVIRONMENTAL JUSTICE	3.0
HASS588	GLOBAL WATER POLITICS AND POLICY	3.0
HASS584	US WATER POLITICS AND POLICY	3.0
HASS593	NATURAL RESOURCES & ENERGY POLICY: THEORIES AND PRACTICE	3.0
MNGN571	ENERGY, NATURAL RESOURCES, AND SOCIETY	3.0
PEGN530	ENVIRONMENTAL LAW AND SUSTAINABILITY	3.0

Leadership

Eric Anderson, Civil & Environmental Engineering, HSE Associate Director

David Benson, HSE Director, Professor, Geology & Geological Engineering

Department of Chemistry

James Ranville, Professor

Bettina Voelker, Professor

Department of Civil & Environmental Engineering

Eric Anderson, Associate Professor

Christopher Higgins, Professor

Terri Hogue, Dean of Earth & Society Programs

Tissa Illangasekare, Professor and AMAX Distinguished Chair

Ning Lu, Professor

Junko Munakata Marr, Professor

John McCray, Professor

John Spear, Professor

Department of Economics and Business

Steven M. Smith, Assistant Professor

Department of Geology and Geological Engineering

Adrienne Marshall, Assistant Professor

Alexis Sitchler, Associate Professor

Danica Roth, Assistant Professor

Kamini Singha, Professor

Paul Santi, Professor

Reed Maxwell, Professor

Wendy Zhou, Professor

Department of Geophysics

John Bradford, Vice President for Global Initiatives

Brandon Dugan, Professor and Baker Hughes Chair in Petrophysics & Borehole Geophysics and Associate Department Head GP

Matthew Siegfried, Assistant Professor

Department of Humanities, Arts and Social Sciences

Hussein Amery, Professor

Adrianne Kroepsch, Assistant Professor

Department of Mechanical Engineering

Nils Tilton, Associate Professor

Department of Mining Engineering

Rennie Kaunda, Assistant Professor

Department of Petroleum Engineering

Yu-Shu Wu, Professor

Materials Science

Degrees Offered

- Master of Science (Materials Science, thesis option or non-thesis option)
- Doctor of Philosophy (Materials Science)

Program Description

The interdisciplinary graduate program in Materials Science exists to educate students, with at least a Bachelor of Science degree in engineering or science in the diverse field of Materials Science. This diversity includes the four key foundational aspects of Materials Science – materials properties including characterization and modeling, materials structures, materials synthesis and processing, and materials performance – as applied to materials of a variety of types (i.e., metals, ceramics, polymers, electronic materials and biomaterials). The Materials Science graduate program is responsible for administering MS (thesis and non-thesis) and PhD degrees in Materials Science.

This interdisciplinary degree program coexists alongside strong disciplinary programs in Chemistry, Chemical and Biochemical Engineering, Mechanical Engineering, Metallurgical and Materials Engineering, Mining, and Physics. The student's graduate committee will have final approval of the course of study.

Fields of Research

- Advanced polymeric materials
- Alloy theory, concurrent design, theory-assisted materials engineering, and electronic structure theory
- Applications of artificial intelligence techniques to materials processing and manufacturing, neural networks for process modeling and sensor data processing, manufacturing process control
- Atomic scale characterization
- Atom Probe Tomography
- Biomaterials
- Ceramic processing, modeling of ceramic processing
- Characterization, thermal stability, and thermal degradation mechanisms of polymers
- Chemical and physical processing of materials, engineered materials, materials synthesis
- Chemical vapor deposition
- Coating materials and applications
- Computational condensed matter physics, semiconductor alloys, first-principles phonon calculations
- Computer modeling and simulation
- Control systems engineering, artificial neural systems for senior data processing, polymer cure monitoring sensors, process monitoring and control for composites manufacturing
- Crystal and molecular structure determination by X-ray crystallography
- Electrodeposition
- Electron and ion microscopy
- Energetic materials (explosives) and processing
- Energy storage
- Experimental condensed-matter physics, thermal and electrical properties of materials, superconductivity, photovoltaics
- Fuel cell materials
- Fullerene synthesis, combustion chemistry
- Heterogeneous catalysis, reformulated and alcohol fuels, surface analysis, electrophotography
- High-temperature ceramics
- Intelligent automated systems, intelligent process control, robotics, artificial neural systems
- Materials synthesis, interfaces, flocculation, fine particles
- Mathematical modeling of material processes
- Mechanical metallurgy, failure analysis, deformation of materials, advanced steel coatings
- Mechanical properties of ceramics and ceramic composites
- High entropy alloys
- Mössbauer spectroscopy, ion implantation, small-angle X-ray scattering, semiconductor defects
- Nano materials
- Nondestructive evaluation
- Nonferrous structural alloys
- Novel separation processes: membranes, catalytic membrane reactors, biopolymer adsorbents for heavy metal remediation of ground surface water
- Numerical modeling of particulate media, thermomechanical analysis
- Optical properties of materials and interfaces
- Phase transformations and mechanisms of microstructural change
- Photovoltaic materials and device processing
- Physical metallurgy, ferrous and nonferrous alloy systems
- Physical vapor deposition, thin films, coatings
- Power electronics, plasma physics, pulsed power, plasma material processing